Chapter 2. PERMUTATIONS

2.3 The Alternating group

Today we will talk about 2-cycles, that is,

TRANSPOSITIONS.

Every permutation can be written as a product of transpositions.

Recall that a transposition

δ is a 2-cycle, so it has the form $\delta = (m\ n)$ (where $m \neq n$) and satisfies $\delta^2 = \varepsilon, \quad \delta^{-1} = \delta$.

In tutorials

THEOREM 2.18. *Every cycle of length $r \geq 1$ is a product of $r-1$ transpositions:* $(k_1\ k_2\ \ldots\ k_r) = (k_1\ k_2)(k_2\ k_3)\ldots(k_{r-2}\ k_{r-1})(k_{r-1}\ k_r)$

Applying
- **the Cycle Decomposition Theorem, and**
- **Theorem 2.18,**

we obtain that

Every permutation is a product of transpositions.

Remark 2.19.

- **Factorisation into transpositions are NOT UNIQUE!**

$$(1\ 2)(2\ 4)(4\ 5) = (1\ 2\ 4\ 5) = (1\ 5)(1\ 4)(1\ 2).$$

- **The numbers of transpositions into any two factorisations have the SAME PARITY.**

Theorem 2.20. Parity Theorem.

If a permutation σ has two factorisations

$$\sigma = \gamma_r \dots \gamma_2\gamma_1 = \mu_s \dots \mu_2\mu_1,$$

where each γ_i and μ_j is a transposition, then both r and s are even or both are odd.

Definitions 2.21. A permutation σ is called **even** or **odd** if it can be written as a product of an even or odd number of transpositions. The set of all even permutations in S_n is denoted A_n, and is called the **alternating group of degree** n.

How can we find the parity of a permutation? And the factorisation into transpositions?

CASE 1: σ **is a cycle**

Then suppose that is an r-cycle. Then the factorisation into transpositions can be obtain from Thm 2.18.

r-cycle = product of r-1 transpositions

r-cycle even (resp. odd) if r-1 is even (resp. odd).

CASE 2: σ is not a cycle

We need to decompose σ into a product of disjoint cycles, and then proceed like in Case 1 to find the parity of each factor cycle.

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 5 & 4 & 6 & 1 & 7 & 8 & 2 & 9 & 3 \end{pmatrix} \in S_9$$

$$\sigma = \underbrace{(1\ 5\ 7\ 2\ 4)}_{\text{5-cycle}}\underbrace{(3\ 6\ 8\ 9)}_{\text{4-cycle}}$$

By Thm 2.18 σ is a product of $4 + 3 = 7$ transpositions, and σ is <u>odd</u>.

$$\sigma = (1\ 5\ 7\ 2\ 4)(3\ 6\ 8\ 9) \overset{\text{Thm 2.18}}{=}$$

$$= (1\ 5)(5\ 7)(7\ 2)(2\ 4)(3\ 6)(6\ 8)(8\ 9)$$